Principal Component Analysis and PCR

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Principal Component Analysis

Principal Component Analysis

- This is the most popular unsupervised procedure ever.
- Invented by Karl Pearson (1901).
- Developed by Harold Hotelling (1933).
- What does it do? It provides a way to visualize high dimensional data, summarizing the most important information.

Principal Components

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  2. as a tool for **data visualization** (unsupervised).
- PCA: process by which principal components are computed.

Visualization

Suppose that we wish to visualize n observations with measurements on a set of p features, $X_1, X_2, ..., X_p$, as part of an exploratory data analysis.

Crime Data

This data-set gives the crime rates per 100k people for the 72 largest cities in the US.

Table 1: Crime Data

	Murder	Rape	Robbery	Assault	Burglary	Larceny	MVT
New-York	21.3	36.3	988.8	814.5	1204.6	2859.9	1300.7
LA	23.8	43.8	868	1123.4	1226.2	3120.5	1434.3
Chicago	33.1	40	1210.5	1440.9	1563.6	4323.4	1427.9
Houston	21.3	53	567.7	665.5	1451.5	3239.2	1287.2
Philly	25.9	46.2	814.2	436.2	903.9	2588.3	1620
San-Diego	9.7	34.5	329	704.8	1102.8	3012	1371.8

Visualization

We could look at all possible $\binom{p}{2}$ scatterplots.

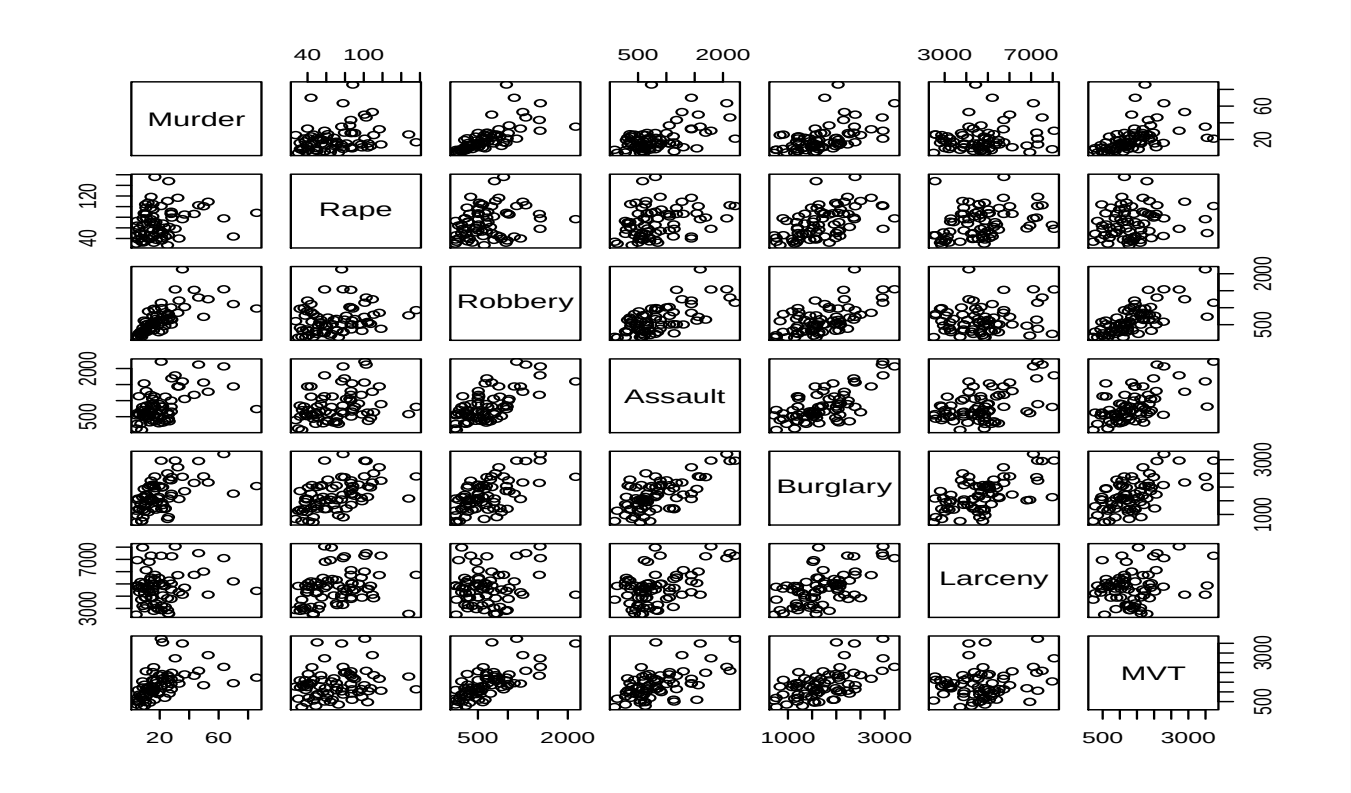


Figure 1: Scatterplot matrix

But ... $\binom{p}{2}$ grows too quickly and each plot has little info.

Projection Methods

- Principal components analysis (PCA) belongs to the class of *projection methods*.
- Such methods choose one or more linear combinations of the original variables to maximize some measure of "interestingness".
- In PCA the goal is to **reduce the dimensionality** of a data set comprised of a **large number of interrelated variables**, while **retaining** as much as possible of the **variation** present in the data.
- How do we do it? A little linear algebra.

Interpreting PCs

- The first PC direction of the data is that along which the observations vary the most.
- If we projected the 100 observations onto this line, then the resulting projected observations would have the **largest possible variance.**

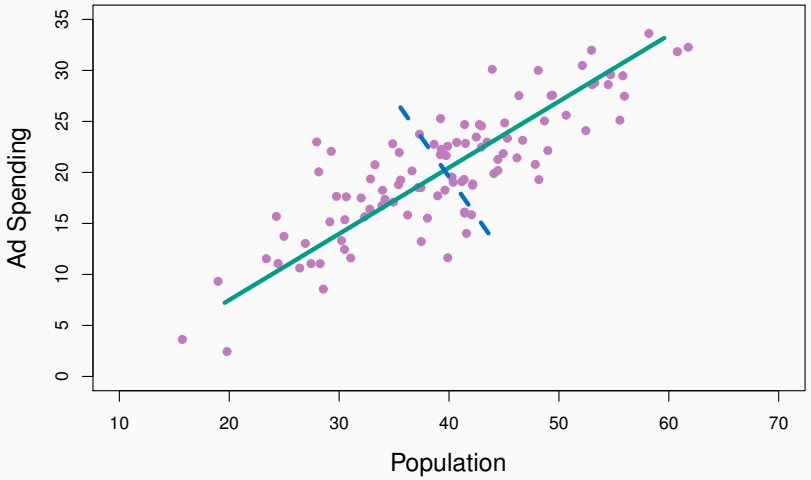


Figure 2: The population size (pop) and ad spending (ad) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

Second Interpretation

- Projecting a point onto a line: *finding the location on the line which is closest to the point.*
- First PC is the direction of the line that is closest to the data points, in terms of squared Euclidean distance.

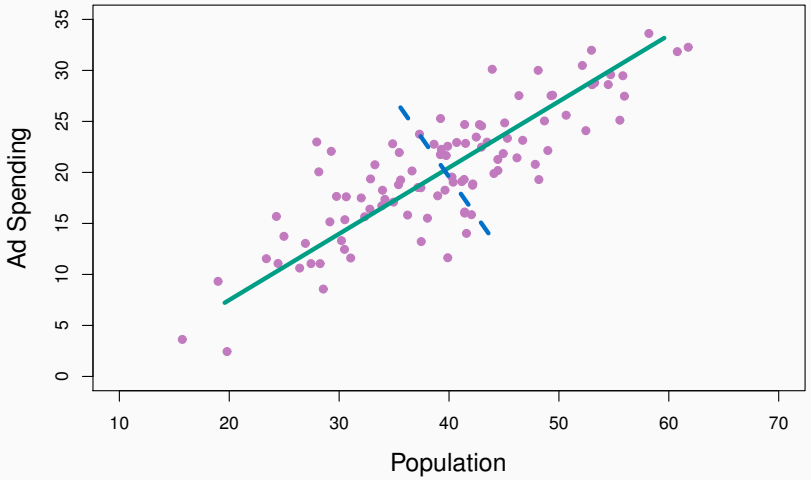
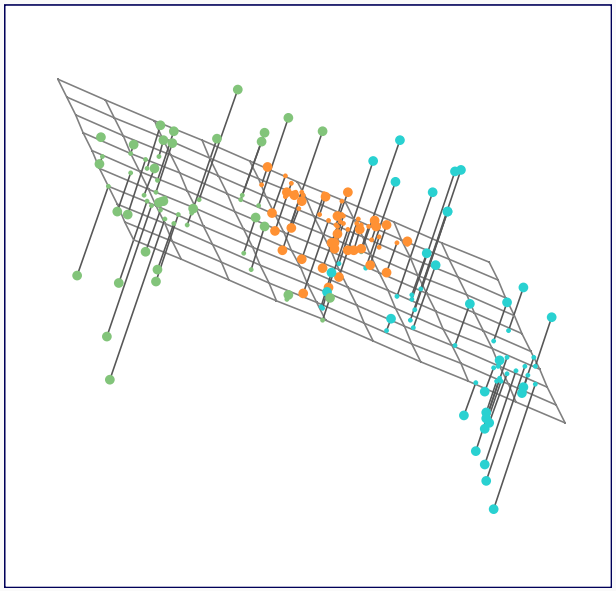
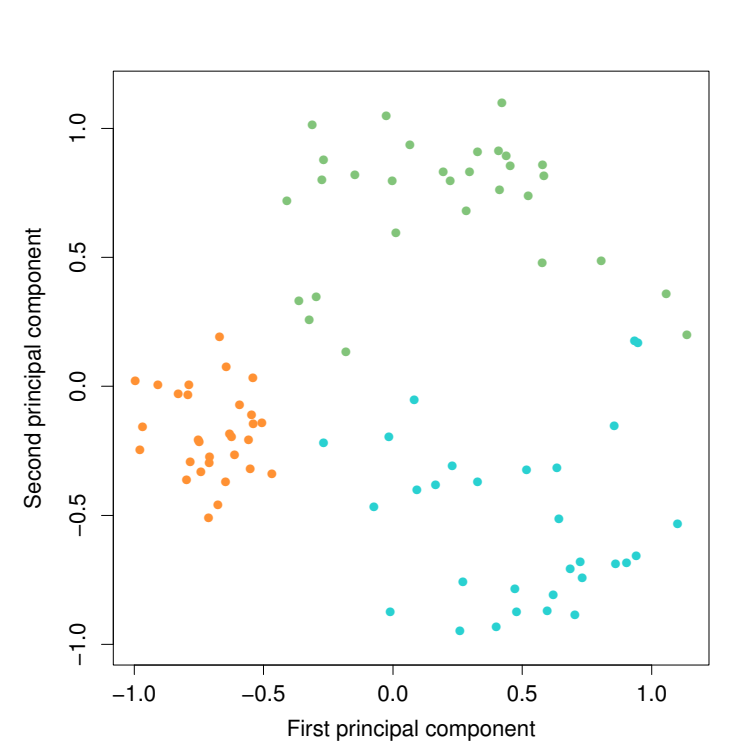


Figure 3: The population size (pop) and ad spending (ad) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

3 Variables

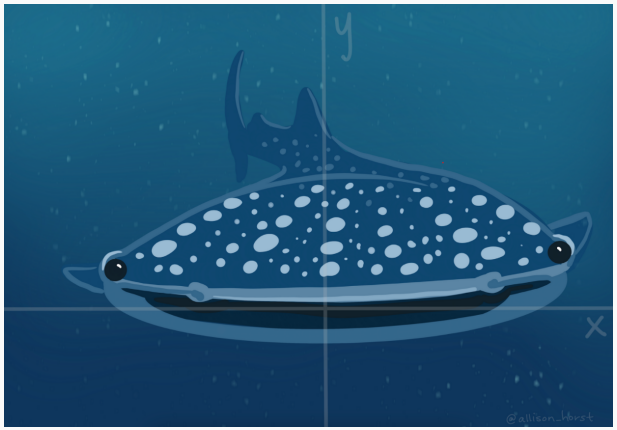
The first two principal components span a plane which is closest to the data.





Intuitive explanation by Allison Horst[1]

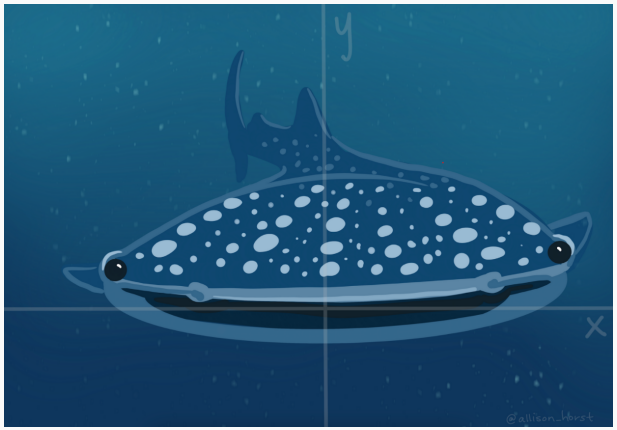
To get as much krill as you can in the fewest passes, you are going to need to change the way you approach the krill swarm!



[1] allisonhorst.com

Intuitive explanation by Allison Horst[2]

To get $\overbrace{\text{as much krill}}^{\text{variance reduction}}$ as you can $\overbrace{\text{in the fewest passes}}^{\text{dimension reduction}}$, you are going to need to $\underbrace{\text{change the way you approach}}_{\text{axis rotation}}$ the krill swarm!.



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Linear Algebra Recap

Eigenvalue [3]

Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say that λ is an eigenvalue and u is the corresponding eigenvector of A if

$$\mathbf{Au} = \lambda \mathbf{u}, \ \mathbf{u} \neq 0$$

Intuitively, this definition means that multiplying A by the vector $\mathbf{u}$ results in a new vector that points in the same direction as u, but is scaled by a factor λ.

[3] from Kevin Murphy's book

"the" eigenvector

Note that, if $\mathbf{u}$ is an eigenvector and c is a scalar, then,

$$\mathbf{A}(c\mathbf{u}) = c(\mathbf{A}\mathbf{u}) = c\lambda\mathbf{u} = \lambda(c\mathbf{u})\ \mathbf{u} \neq 0$$

Hence $c\mathbf{u}$ is also an eigenvector. For this reason when we talk about "the" eigenvector associated with λ, we usually assume that the eigenvector is normalized to have length 1.

This still creates some ambiguity, since $\mathbf{u}$ and $-\mathbf{u}$ will both be eigenvectors, but we will have to live with this

Characteristic equation

We can rewrite the equation above to state that $(\lambda, \mathbf{u})$ is an eigenvalue-eigenvector pair of A if

$$(\lambda \mathbf{I} - \mathbf{A})\mathbf{u} = 0, \ \mathbf{u} \neq 0$$

This equation has a non-zero solution for $\mathbf{u}$ if and only if the matrix $(\lambda \mathbf{I} - \mathbf{A})$ is singular which implies

$$\det(\lambda \mathbf{I} - \mathbf{A}) = 0 \text{ characteristic equation for } A$$

The n solutions of this equation are the n (possibly complex-valued) eigenvalues λ_i, and u_i are the corresponding eigenvectors.

It is standard to sort the eigenvectors in order of their eigenvalues, with the largest magnitude ones first.

Properties

1. The trace of a matrix is equal to the sum of its eigenvalues:

$$tr(\mathbf{A}) \doteq sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i.$$

2. The determinant of a matrix is equal to the product of its eigenvalues:

$$|\mathbf{A}| = \prod_{i=1}^{n} \lambda_i.$$

3. The rank of a matrix A is the number of non-zero eigenvalues of $\mathbf{A}$.
4. If $\mathbf{A}$ is invertible, i.e. if $\mathbf{A}^{-1}$ exists, then λ_i^{-1} will be the eigenvalue of $\mathbf{A}^{-1}$ associated with the eigenvector $\mathbf{u}_i$.

$$\mathbf{A}^{-1}\mathbf{u}_i = (1/\lambda_i)\mathbf{u}_i.$$

5. The eigenvalues of a diagonal (or triangular) matrix are just the diagonal entries.

Mathematics of PCA

Mathematics of PCA

- Let $\mathbf{x}$ be a $p \times 1$ vector and $\text{Cov}(\mathbf{x}) = \Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T]$.
- Interested in reducing the dimensionality p of the original variables, by considering linear combinations of them and retaining the "most important" of these new variables.
- Define a new variable: $z_1 = \beta_1' \mathbf{x} = \sum_{j=1}^{p} \beta_{j1} x_j$.
- How to choose the weights β_1?
- For PCA, we want to maximize the variance of the new variable z_1.

First PC

- Criterion:
  $\max_{z_1} \mathrm{Var}(z_1) = \max \mathrm{Var}(\sum_{j=1}^{p} \beta_{j1} x_j) = \max \beta_1' \Sigma \beta_1$.
- Easy to see that the maximum would be achieved for an infinite β_1; thus, a normalization constraint must be imposed.
- Standard constraint: $||\beta_1|| = 1$.
- New Criterion: $\max \beta_1' \Sigma \beta_1$ s.t. $||\beta_1||^2 = 1$ [4].
- Rayleigh-Rietz theorem: the solution is given by the eigenvector corresponding to the largest eigenvalue λ_1 of Σ.

[4] $||x||^2 = \sum_{i=1}^{n} x_i^2$

Second PC

- If we want a second linear combination: $z_2 = \beta_2' \mathbf{x}$, then the problem becomes:
- Criterion: $\max \beta_2' \Sigma \beta_2$ s.t. $||\beta_2||^2 = 1$, and $\langle \beta_1, \beta_2 \rangle = 0$.
- In general, we can construct p linear combinations $\mathbf{z} = \mathbf{B}'\mathbf{x}$, $\mathbf{B}$ being a $p \times p$ matrix whose $k - th$-column corresponds to β_k.
- The optimal set of weights is given by the eigenvectors of Σ.

$$\Sigma = B \Lambda B' \text{ where } \Lambda = Diag(\lambda_1, \ldots, \lambda_p).$$

- The linear combinations $z_1, z_2, \ldots$ are called the principal components (or scores) and the weight vectors $\beta_1, \beta_2, \ldots$ are called the loading vectors.
- *Principal Components of y are by construction an orthonormal linear transformation of x.*

Linear Algebra

- Transformation: p linear combinations $\mathbf{z} = \mathbf{B}'\mathbf{x}$.
- The principal components of $\mathbf{x}$ are the eigen-vectors $B = [\beta_1; \beta_2; \cdots; \beta_p]$ corresponding to eigen values $\Lambda = \text{Diag}(\lambda_1, \ldots, \lambda_p)$.

$$\Sigma = B\Lambda B' \text{ where } \Lambda = Diag(\lambda_1, \ldots, \lambda_p) \tag{1}$$

$$\Rightarrow \Sigma B = B\Lambda \text{ since } B'B = I \text{ when } B \text{ is orthogonal} \tag{2}$$

$$\Rightarrow \mathbf{V}(z) = B'\Sigma B = B'(B\Lambda) = \Lambda \tag{3}$$

- In English: Hence, the principal components $Z_1, Z_2, \ldots, Z_p$ are uncorrelated and $\mathbf{V}(Z_j) = \lambda_j$.

Proportion of Variance Explained

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$$\text{Total Variance } = \sum_{j=1}^{p} \sigma_j^2 = trace(\Sigma) = trace(B\Lambda B')$$

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- So, $\sum_{j=1}^{p} \mathbf{V}(Z_j) = \sum_{j=1}^{p} \mathbf{V}(X_j)$, and each PC "explains" $\lambda_j / \sum_{j=1}^{p} \lambda_j$ proportion of variance.

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- For a high-dim data, at some point the proportions will fall.
- We can throw out those components that do not contribute much to the total variance.

PCA applied to data

- In practice, we observe n observations of the $(x_1, \ldots, x_p)$.
- Let X be a data matrix with n rows and p columns. Subtract the column mean from each variable.
- The first principal component direction is the vector $\beta_1 = (\beta_{11}, \ldots, \beta_{p1})'$ s.t.

$$\max_{\beta_{11},\ldots,\beta_{1p}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \left(\sum_{j=1}^{p} \beta_{j1} x_{ij} \right)^2 \right\} \text{ subject to } \sum_{j=1}^{p} \beta_{j1}^2 = 1 \quad (4)$$

- Projection of ith sample into β_1 - also called **score** y_{i1}.
- PC directions $\beta_1 = (\beta_{11}, \ldots, \beta_{p1})'$ - also called **loadings**.

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- Maximize sample variance of the scores y_{i1}.

Example 1: Crime Data

- The first example is the US Crime data that give crime rates per 100,000 people for the 72 largest US cities in 1994. There are 7 variables: 1) Murder, 2) Rape, 3) Robbery, 4) Assault, 5) Burglary, 6) Larceny, 7) Motor Vehicle Thefts.
- **We first center and scale the data.**
- Then apply principal components using R package `princomp`.
- The first figure plots the first two principal components of these data. The figure represents both the principal component **scores** and the **loading vectors** in a single **biplot** display.

Example 1: Crime Data

Table 2: Principal Components

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7
Std. Dev.	1.93	1.08	0.87	0.70	0.57	0.45	0.42
Prop. Var	0.54	0.17	0.10	0.07	0.04	0.03	0.02
Cumu Prop.	**0.54**	**0.71**	**0.82**	0.89	0.94	0.97	1

The first two PC's explain 71.3% of the total variance, the first three PC's together explain 82.3% of the total variance, and so on.

What are the principal components?

- Recall, the coefficients for the principal components are called loadings, i.e. if $z_1 = \beta_1' \mathbf{x}$ is the first PC, β_1 is the corresponding loading vector.
- The loading vector β_1 with elements $\beta_{11}, \ldots, \beta_{p1}$ defines a direction in feature space along which the data vary the most.
- Constraining $\mathbf{z}_1$ to be uncorrelated with $\mathbf{z}_2 \equiv$ the directions β_1 and β_2 are orthogonal.

Table 3: Loadings β

Loadings:							
	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7
Murder	**-0.37**	**-0.339**	0.202	0.717	0.277	0.22	-0.262
Rape	**-0.249**	**0.467**	0.783	-0.159		0.267	
Robbery	**-0.426**	**-0.388**			-0.194	-0.147	0.776
Assault	**-0.434**		-0.282		-0.767	0.118	-0.358
Burglary	**-0.45**	**0.238**			0.242	-0.794	-0.22
Larceny	**-0.276**	**0.606**	-0.492	0.209	0.264	0.303	0.331
MVT	**-0.39**	**-0.303**	-0.134	-0.644	0.399	0.351	-0.204

Biplot

Biplot is a nice visualization tool. It shows the outcome of a PCA by superimposing two scatterplots on one figure:

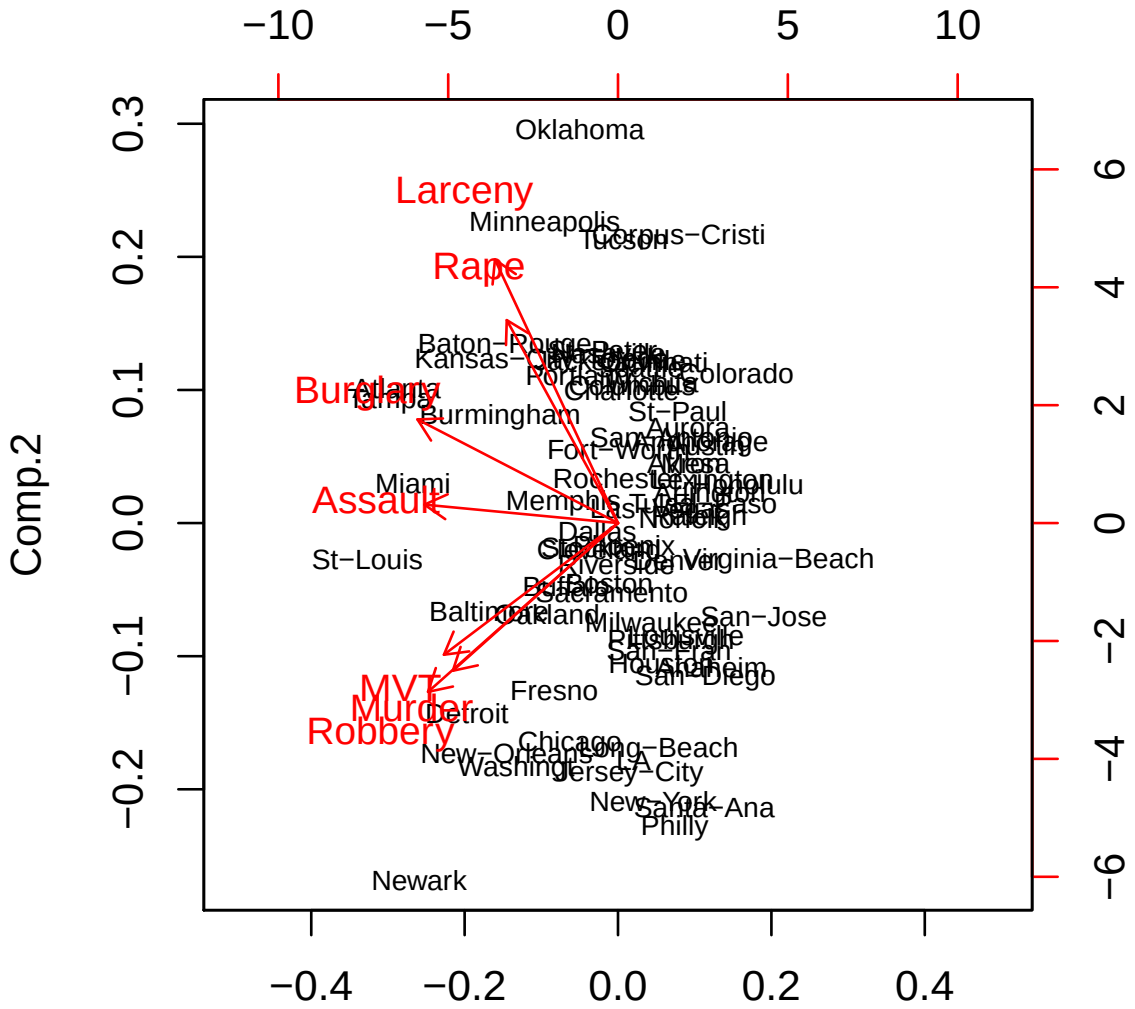
- the score of each case (i.e., 72 cities) on the first two principal components.

$$z_{i1} = \sum_{j=1}^{p} \beta_{j1} x_{ij}, i = 1, \ldots, n. \tag{6}$$

$$z_{i2} = \sum_{j=1}^{p} \beta_{j2} x_{ij}, i = 1, \ldots, n. \tag{7}$$

- the loading of each variable (i.e., each 7 crime categories) on the first two principal components ($\beta_{j1}, \beta_{j2}, j = 1, \ldots, p.$).

Biplot


−10
−5
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0.3
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Comp.2
6
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2
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−2
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Oklahoma
Larceny
Minneapolis
Corpus−Cristi
Tucson
Rape
Baton−Rouge
Kansas−City
Colorado
Burglary
Atlanta
Tampa
Burmingham
St−Paul
Aurora
Fort−Worth
Rochester
Miami
Assault
Memphis
Dallas
St−Louis
Virginia−Beach
Sacramento
Baltimore
Oakland
Milwaukee
San−Jose
San−Diego
MVT
Murder
Robbery
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LA
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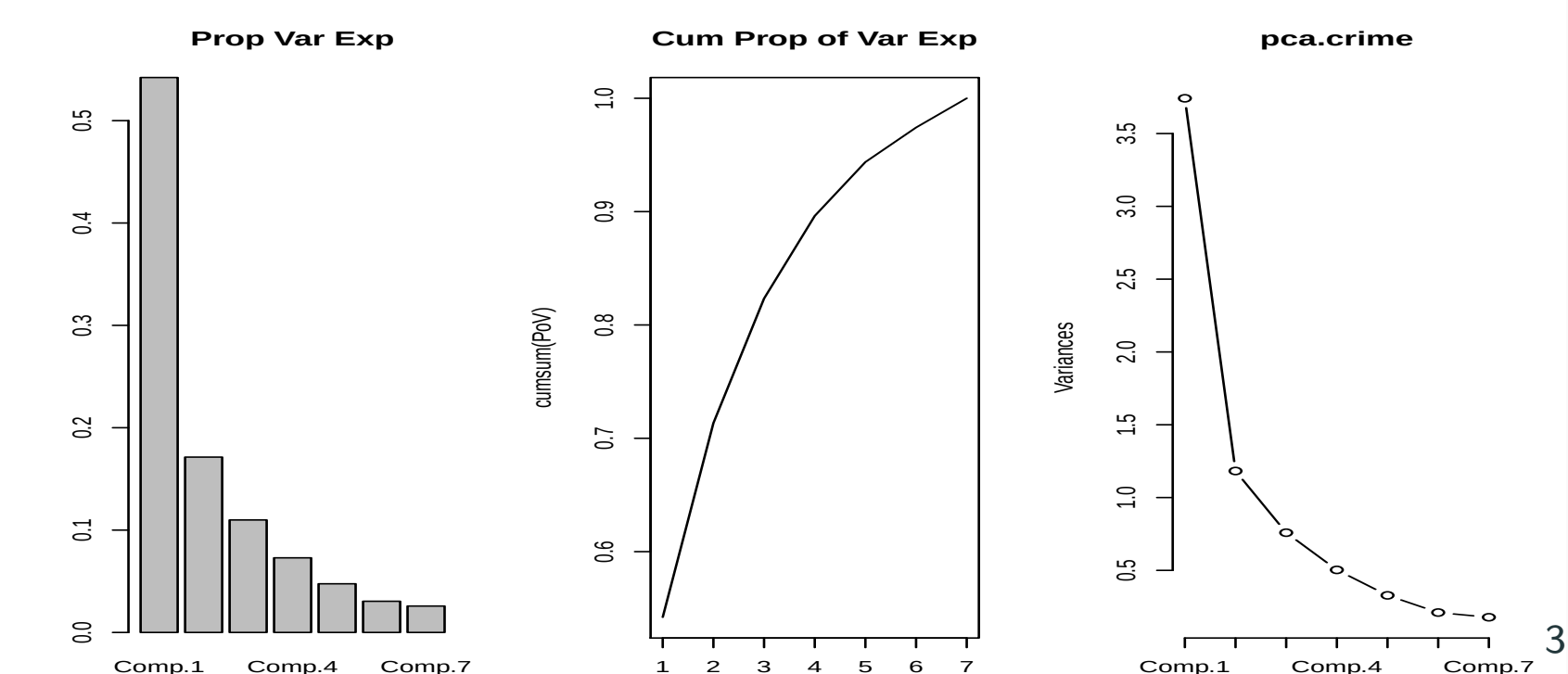
- The second PC distinguishes between cities with high rape and larceny incidents (and to some degree burglaries) and cities with high murder, robbery and MVT incidents.
- Be very careful with such interpretations. This is data from 1994.

How many?

How many principal components are needed? Unfortunately, there is no single answer. We typically decide on the number of PCs required to visualize the data by eyeballing the scree plot, and looking for a point at which the proportion of variance explained by each subsequent PC drops off (elbow point).

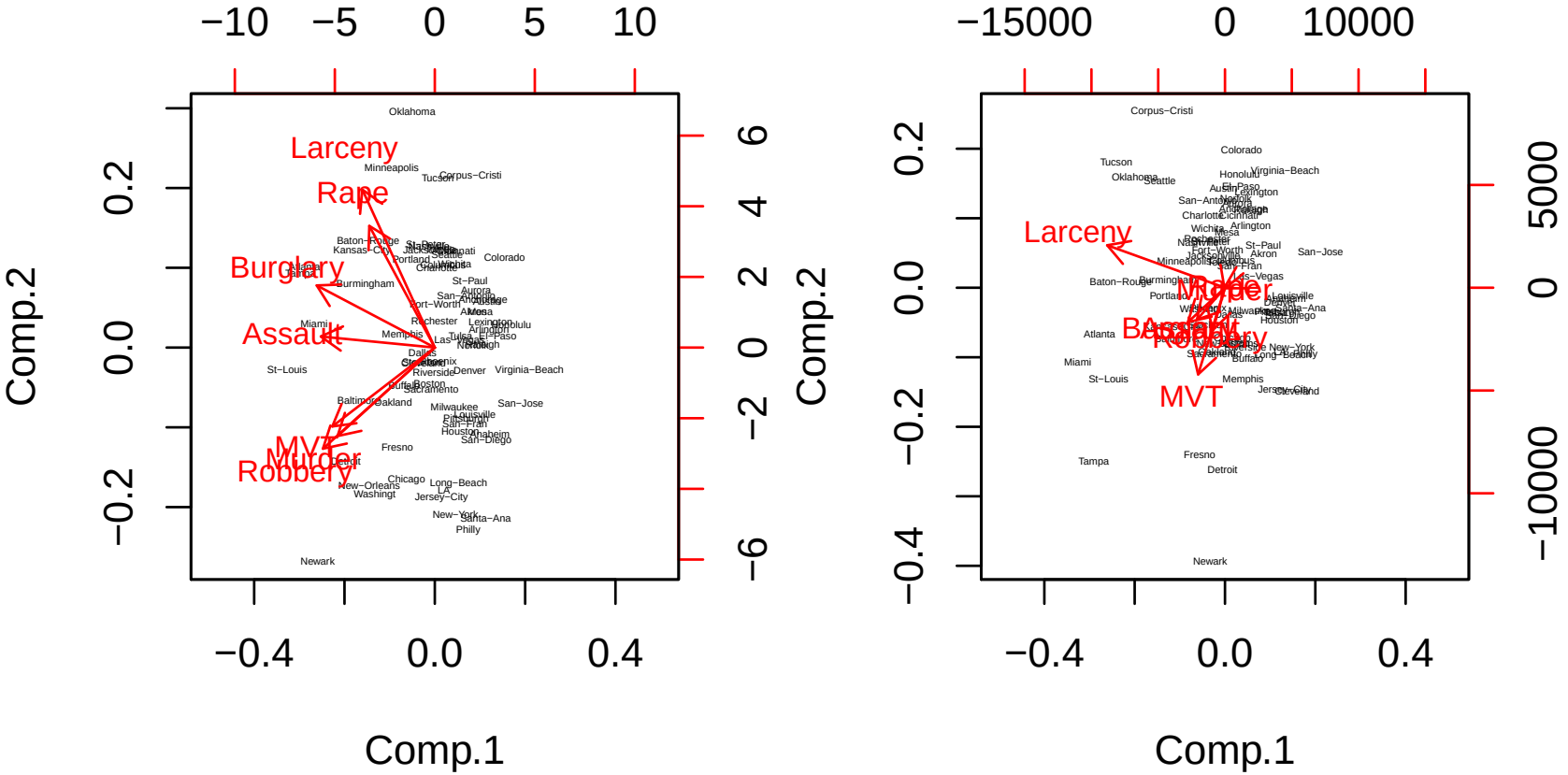


Scaling the variable

Most of the time, we don't care about the absolute numerical value of a variable. We care about the value relative to the spread observed in the sample.

In that case, before PCA, in addition to centering each variable, we also multiply it times a constant to make its variance equal to 1, i.e. we standardize each variable.

Scaled versus Unscaled



PCA for Supervised Learning

- We can also use PCA for supervised learning.
- PC Regression: Use PC's instead of the variables.
- Many statistical techniques, such as regression, classification, and clustering, can be easily adapted to use the $n \times M$ matrix whose columns are the first $M \ll p$ principal component score vectors, rather than using the full $n \times p$ data matrix.
- Less noisy results, since it is often the case that the signal (as opposed to the noise) in a data set is concentrated in its first few PCs.

Generalization

PCA works under a Euclidean geometry in the space of variables.

Often, the natural geometry is different:

- We expect some variables to be "closer" to each other than to other variables.
- Some correlations between variables would be more surprising than others.

Examples:

1. Variables are pixel values, samples are different images of the brain. We expect neighboring pixels to have stronger correlations.
2. Variables are rainfall measurements at different regions. We expect neighboring regions to have higher correlations.

Excellent Reference: Susan Holmes. Multivariate Analysis, the French way. (2006).

Extension: Principal Components Regression

PCR

PCR is a popular regression technique that uses the principal components as the covariates or predictors in a linear regression model.
We will need a quick recap of PCA and multiple regression before we can talk about PCR.

Quick recap

- First PC: $\max_{z_1} \mathrm{Var}(z_1) = \max \mathrm{Var}(\sum_{j=1}^{p} \beta_{j1} x_j) = \max \beta_1' \Sigma \beta_1$ subject to the constraint: $||\beta_1|| = 1$.
- Rayleigh-Rietz theorem: the solution is given by the eigenvector corresponding to the largest eigenvalue λ_1 of Σ.
- Second PC: $\max \beta_2' \Sigma \beta_2$ s.t. $||\beta_2||^2 = 1$, and $\langle \beta_1, \beta_2 \rangle = 0$.
- In general, we can construct p linear combinations $\mathbf{z} = \mathbf{B}'\mathbf{x}$, $\mathbf{B}$ being a $p \times p$ matrix whose $k - th$-column corresponds to β_k.
- The optimal set of weights is given by the eigenvectors of Σ.

$$\Sigma = B \Lambda B' \text{ where } \Lambda = Diag(\lambda_1, \ldots, \lambda_p).$$

- The linear combinations $z_1, z_2, \ldots$ are called the principal components (or scores) and the weight vectors $\beta_1, \beta_2, \ldots$ are called the loading vectors.

Notations

- In the following section, we will slightly change our notations and denote the eigenvectors as $\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_p$.
- So to rewrite this part:
- The optimal set of weights is given by the eigenvectors of Σ.

$$\Sigma = \Phi \Lambda \Phi^T \text{ where } \Lambda = Diag(\lambda_1, \ldots, \lambda_p).$$

- The linear combinations $z_1, z_2, \ldots$ are called the principal components (or scores) and the weight vectors $\boldsymbol{\phi}_1, \boldsymbol{\phi}_2, \ldots$ are called the loading vectors.
- You'll see we'll use β in a way that resembles OLS in the final form.

Dimension reduction methods

- The methods that we have discussed so far in this chapter have involved fitting linear regression models, via least squares or a shrunken approach, using the original predictors, $X_1, X_2, \ldots, X_p$.
- We now explore a class of approaches that transform the predictors and then fit a least squares model using the transformed variables. We will refer to these techniques as dimension reduction methods.

Dimension reduction – details

- Let $Z_1, Z_2, \ldots, Z_M$ represent $M < p$ *linear combinations* of our original p predictors. That is,

$$Z_m = \sum_{j=1}^{p} \phi_{mj} X_j \tag{1}$$

  for some constants $\phi_{m1}, \ldots, \phi_{mp}$.
- We can then fit the linear regression model,

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \epsilon_i, \quad i = 1, \ldots, n, \tag{2}$$

  using ordinary least squares.
- Note that in model (2), the regression coefficients are given by $\theta_0, \theta_1, \ldots, \theta_M$. If the constants $\phi_{m1}, \ldots, \phi_{mp}$ are chosen wisely, then such dimension reduction approaches can often outperform OLS regression.

Dimension reduction – details ... continued

- Notice that from definition (1),

$$\sum_{m=1}^{M} \theta_m z_{im} = \sum_{m=1}^{M} \theta_m \sum_{j=1}^{p} \phi_{mj} x_{ij} = \sum_{j=1}^{p} \sum_{m=1}^{M} \theta_m \phi_{mj} x_{ij} = \sum_{j=1}^{p} \beta_j x_{ij},$$

  where

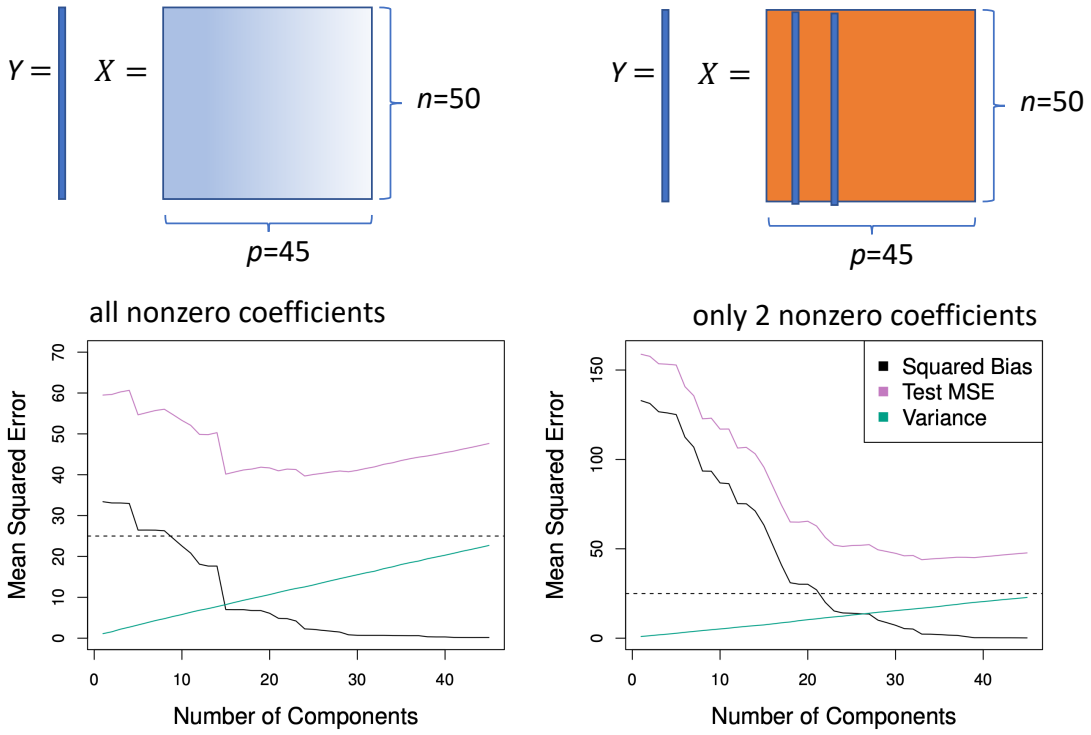
$$\beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj}. \tag{3}$$

- Hence model (2) can be thought of as a special case of the original linear regression model.
- Dimension reduction serves to constrain the estimated β_j coefficients, since now they must take the form (3).
- Can win in the bias-variance tradeoff.

Principal components regression

- Here we apply principal components analysis (PCA) (discussed in Chapter 10 of ISLR) to define the linear combinations of the predictors, for use in our regression.
- The first principal component is that (normalized) linear combination of the variables with the largest variance.
- The second principal component has largest variance, subject to being uncorrelated with the first.
- And so on.
- Hence with many correlated original variables, we replace them with a small set of principal components that capture their joint variation.

PC Regression applied to synthetic datasets



PCR was applied to two simulated data sets. The black, green, and purple lines correspond to squared bias, variance, and test mean squared error, respectively.

Partial Least Squares

- PCR identifies linear combinations, or directions, that best represent the predictors $X_1, \ldots, X_p$.
- These directions are identified in an unsupervised way, since the response Y is not used to help determine the principal component directions.
- That is, the response does not supervise the identification of the principal components.
- Consequently, PCR suffers from a potentially serious drawback: there is no guarantee that the directions that best explain the predictors will also be the best directions to use for predicting the response.

Partial Least Squares … continued

- Like PCR, PLS is a dimension reduction method, which first identifies a new set of features $Z_1,\ldots,Z_M$ that are linear combinations of the original features, and then fits a linear model via OLS using these M new features.
- But unlike PCR, PLS identifies these new features in a supervised way – that is, it makes use of the response Y in order to identify new features that not only approximate the old features well, but also that are related to the response.
- Roughly speaking, the PLS approach attempts to find directions that help explain both the response and the predictors.

Details of partial least squares

- After standardizing the p predictors, PLS computes the first direction Z_1 by setting each ϕ_{1j} in (1) equal to the coefficient from the simple linear regression of Y onto X_j.
- One can show that this coefficient is proportional to the correlation between Y and X_j.
- Hence, in computing $Z_1 = \sum_{j=1}^{p} \phi_{1j} X_j$, PLS places the highest weight on the variables that are most strongly related to the response.
- Subsequent directions are found by taking residuals and then repeating the above prescription.